

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Space Management Tool
<i>Company:</i>	RobotDay
Contact Details	
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<i>Additional Contact:</i>	Additional contact name
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Additional Info	
<i>Students must sign NDA:</i>	☒
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Click or tap here to enter text.
Project Background Information	
Robot Day needs your help to inspire more people in more places about the wonder of robotics, automation and AI. Work with a dedicated team of volunteers with a wealth of engineering, industry and STEAM experience to build a modern event layout tool that will help us scale beyond labour-intensive manual planning of events to become an organisation with global potential. First Nottingham, then the world? Robot Day is a celebration of Science, Technology, Engineering, Arts and Maths (STEAM) for people of all ages. Our events showcase the value of robotics to society and raise public engagement with issues relating to robotics, automation and artificial intelligence (AI). Robot Days are free events aimed at people of all ages and backgrounds. Visitors to a typical event have the chance to build their own robot, get hands-on with coding, hear from inspirational experts about how robots improve people's lives, and experience virtual reality. There are also artistic performances and exhibits that explore the themes through creative works such as craft and dance workshops and a range of static and moving exhibits. www.robotday.co.uk/about Robot Day Coventry is a non-profit organisation run by volunteers, who want to expand our reach and bring events to more cities, including Nottingham, in	

¹ An overview of different research areas in the school of computer science can be found [here](#).

partnership with the Nottingham University Computer Science department. Previously we also held events in Derby, but due to limited administrative resources, the volunteer team is now only able to organise one event per year.

One of the most time-consuming aspects of organising the event is laying out the spaces with exhibits, workshops, demos to meet the requirements of contributors. Help us build a modern, scalable event layout system that will help us and our partners to run events in many locations with confidence that no exhibitor requirements are missed.

Note about NDA: We want you to be able to talk about our project and view this as important for your future. A limited NDA is required to protect personal data, inventive steps and other specific aspects of the project. We will develop an NDA that gives you the maximum possible opportunity to discuss your work.

Requirements

Minimum viable product

1. *Must be a 2D layout tool*
2. *Must be compliant with existing spatial ontology*
3. *Must import building data from plans (including where possible restrictions on the space e.g. max. person count, door heights, socket locations)*
4. *Must import exhibitor data from planning spreadsheet*
5. *Must automatically generate exhibit "blocks" that can be dragged onto the building plans and moved around to define layout (e.g. to represent tables, chairs, other equipment)*
6. *Must auto-detect clashes*
7. *Must output exhibition map (the imported plan plus overlay of exhibit locations)*
8. *Must be operable by non-expert users*

Stretch tasks

1. *Should auto-generate directional signs and door labels*
2. *Should generate formatted/annotated exhibition guide*
3. *Should give automated layout suggestions*
4. *Could perform layout optimisation*
5. *Could provide visitor flow analysis and optimisation*
6. *Could perform allocation of helper volunteers to contributors/exhibition spaces*
7. *Could include a 3D representation of the space*

Constraints

1. *Has to work with data formats as defined by the spreadsheets provided*
2. *Has to be GDPR compliant*
3. *Should avoid any external subscription-based services*
4. *Should follow open-source principles*
5. *Has to have a maintainable codebase*

Skills	
<i>Highly desirable: Interest in user interface design and implementation</i>	<i>Desirable:</i>
Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.